

Profiling

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1 Profiling

1.1 Setup

Let us load the data and libraries.

```
library(stats)
library(rpart)
library(rpart.plot)
library(corrplot)
library(car)
library(vioplot)

## Loading required package: sm
## Package 'sm', version 2.2-6.0: type help(sm) for summary information
## Loading required package: zoo
##
## Attaching package: 'zoo'
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric

library(effsize)

source(here::here("R", "constants.R"))
source(here::here("R", "load_data.R"))

data <- load_bank_data()
logistic_regression <- readRDS(here::here("models", "log_regression.rds"))
decision_tree <- readRDS(here::here("models", "decision_tree.rds"))
head(data)
```

```
##   age      job marital education default balance housing loan  contact day
## 1  30 unemployed married  primary      no    1787      no   no cellular  19
## 2  33   services married secondary     no    4789     yes yes cellular  11
## 3  35 management single  tertiary     no    1350     yes  no cellular  16
## 4  30 management married  tertiary     no    1476     yes yes unknown   3
## 5  59 blue-collar married  secondary     no      0     yes  no unknown   5
## 6  35 management single  tertiary     no     747      no   no cellular  23
##   month duration campaign pdays previous poutcome y
## 1  oct       79         1    -1         0 unknown no
## 2  may      220         1   339         4 failure no
## 3  apr      185         1   330         1 failure no
## 4  jun      199         4    -1         0 unknown no
## 5  may      226         1    -1         0 unknown no
## 6  feb      141         2   176         3 failure no
```

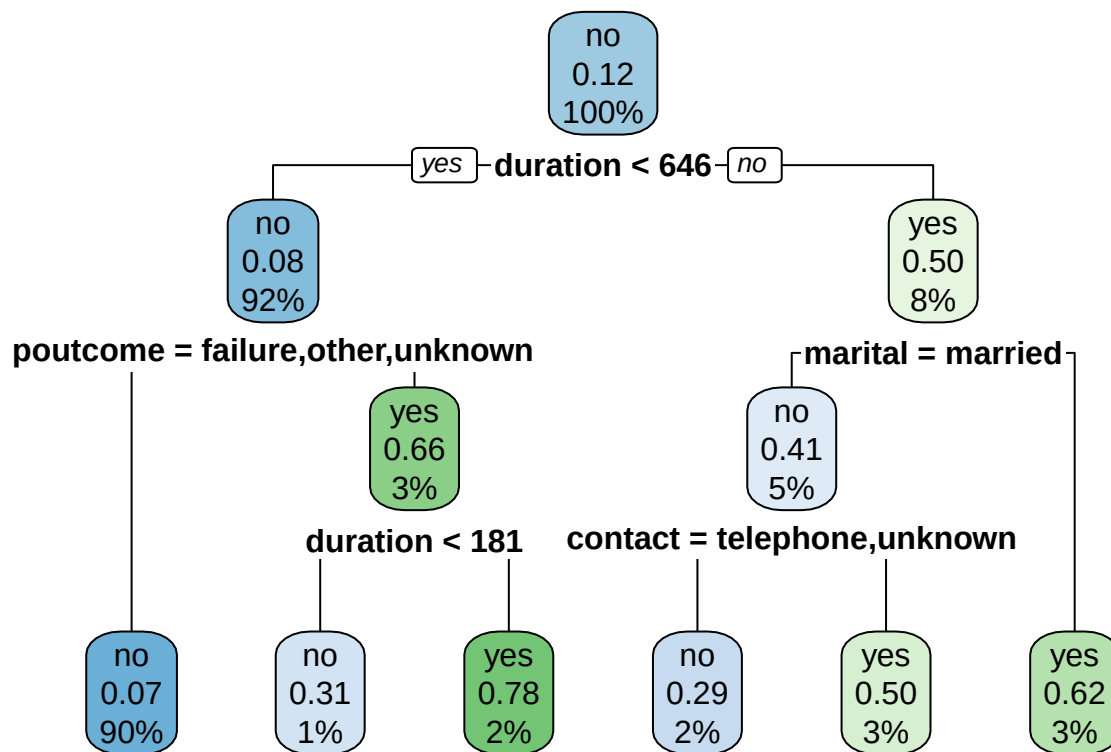
We will also transform the data the same way that we have done during modeling.

```
data.joint_job <- data
data.joint_job[!(data$job == 'blue-collar' | data$job == 'retired'), "job"] <- "unknown"
data.pos <- data[data$y == 'yes',]
data.joint_job.pos <- data.joint_job[data.joint_job$y == 'yes',]

summary(logistic_regression)
```

```
##
## Call:
## glm(formula = y ~ job + marital + housing + loan + contact +
##      poutcome + I(duration^lambda), family = binomial, data = data.specific_job[-569,
##      ])
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.4234144  0.2887112 -11.858  < 2e-16 ***
## jobretired      1.1079931  0.2505248   4.423  9.75e-06 ***
## jobunknown      0.4337103  0.1636434   2.650  0.00804 **
## maritalmarried -0.4286682  0.1684699  -2.544  0.01094 *
## maritalsingle  -0.1144606  0.1829358  -0.626  0.53152
## housingyes     -0.5092094  0.1200770  -4.241  2.23e-05 ***
## loanyes        -0.7889140  0.1921775  -4.105  4.04e-05 ***
## contacttelephone -0.0867686  0.2201306  -0.394  0.69346
## contactunknown -1.1175292  0.1742463  -6.414  1.42e-10 ***
## poutcomeother   0.4754379  0.2587739   1.837  0.06617 .
## poutcomesuccess  2.3901548  0.2561809   9.330  < 2e-16 ***
## poutcomeunknown -0.3070400  0.1725872  -1.779  0.07523 .
## I(duration^lambda) 0.0188705  0.0008714  21.656  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 3230.8  on 4519  degrees of freedom
## Residual deviance: 2256.2  on 4507  degrees of freedom
## AIC: 2282.2
##
## Number of Fisher Scoring iterations: 6
```

```
rpart.plot(decision_tree)
```



1.1.1 Data segmentation

In this subsection we will segment the dataset into three parts, based on logistic regression predictions. The predicted probability lying in interval (0, 0.3) will correspond to *Low*, (0.3, 0.7) to *Medium* and (0.7, 1.0) to *High* segments.

```

plot_count_bars <- function(col, arg_data, ...) {
  pos_labeled = arg_data[arg_data$y == "yes",]
  neg_labeled = arg_data[arg_data$y == "no",]
  pos <- aggregate(x = list(count = pos_labeled$y), by = list(column = pos_labeled[,col]), FUN = length)
  neg <- aggregate(x = list(count = neg_labeled$y), by = list(column = neg_labeled[,col]), FUN = length)
  data_matrix <- matrix(
    c( pos$count, neg$count), nrow = 2, byrow = TRUE
  )
  colnames(data_matrix) <- pos$column
  rownames(data_matrix) <- c("Pos. Cnt", "Neg. Cnt")
  barplot(
    data_matrix,
    beside = TRUE,
    legend = rownames(data_matrix),
    main = sprintf("Class Counts X %s", col),
    xlab = col, ylab = "Class Counts",
    ...
  )
}

```

```

    grid()
}

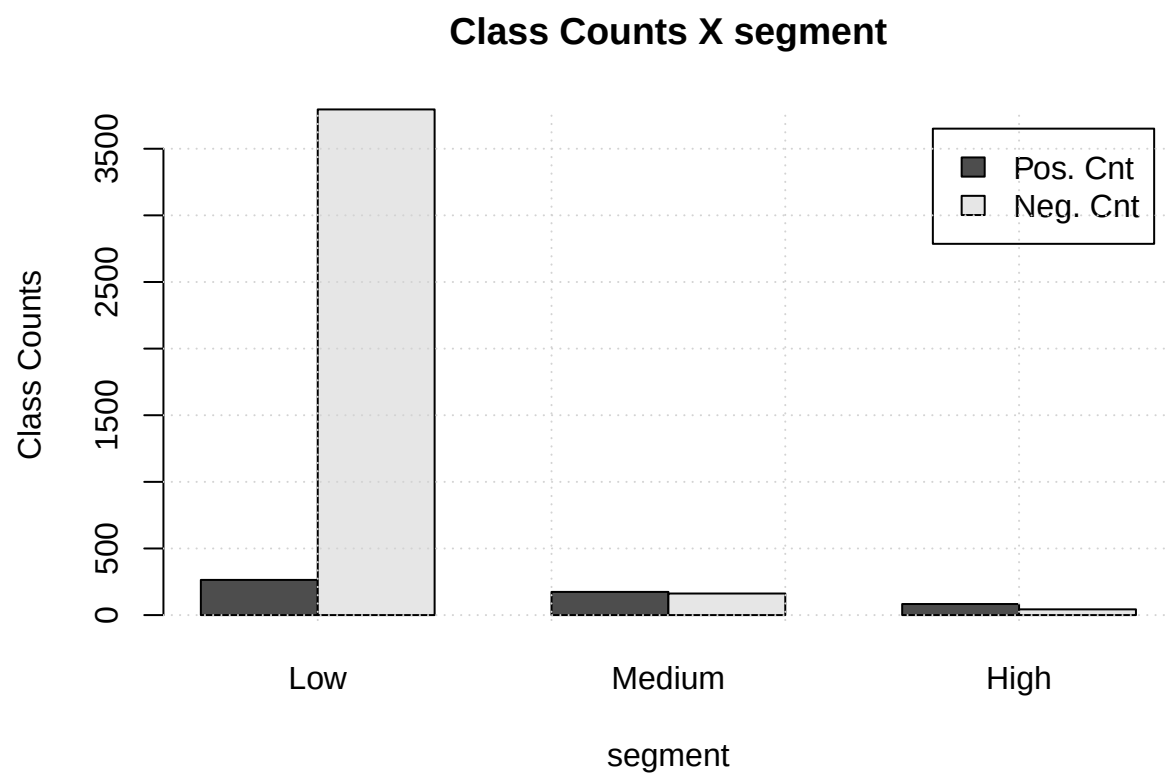
plot_rateBars <- function(col, arg_data, ...) {
  pos_labeled = arg_data[arg_data$y == "yes",]
  neg_labeled = arg_data[arg_data$y == "no",]
  par(mar=c(5, 4, 6, 2))
  total_cnt <- aggregate(x = list(count = arg_data$y), by = list(column = arg_data[,col]), FUN = length)
  pos <- aggregate(x = list(count = pos_labeled$y), by = list(column = pos_labeled[,col]), FUN = length)
  neg <- aggregate(x = list(count = neg_labeled$y), by = list(column = neg_labeled[,col]), FUN = length)
  data_matrix <- matrix(
    c( pos$count / total_cnt, neg$count / total_cnt), nrow = 2, byrow = TRUE
  )
  colnames(data_matrix) <- pos$column
  rownames(data_matrix) <- c("Pos. Rate", "Neg. Rate")
  barplot(
    data_matrix,
    main = sprintf("Class Rates X %s", col),
    ylim=c(0,1),
    xlab = col, ylab = "Class Rates",
    ...
  )
  grid()
  legend(
    "topright",
    legend = rownames(data_matrix),
    xpd = T,
    fill=c("black", "grey"),
    inset=c(0, -0.2)
  )
}

data.joint_job$score <- predict(logistic_regression, newdata=data.joint_job, type='response')
data.joint_job$segment <- cut(data.joint_job$score, breaks = c(-Inf, 0.3, 0.7, Inf), labels = c("Low", "Mid", "High"))

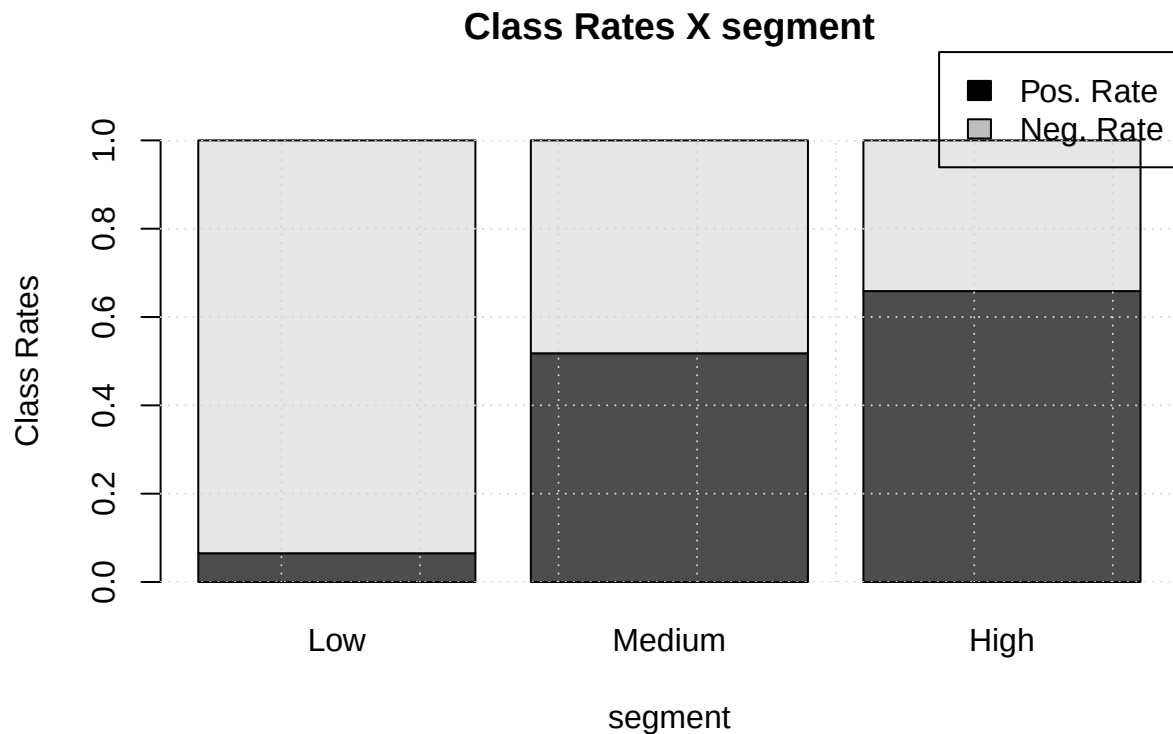
data.joint_job.pos$score <- predict(logistic_regression, newdata = data.joint_job.pos, type='response')
data.joint_job.pos$segment <- cut(data.joint_job.pos$score, breaks = c(-Inf, 0.3, 0.7, Inf), labels = c("Low", "Mid", "High"))

plot_countBars("segment", data.joint_job)

```



```
plot_rate_bars( "segment", data.joint_job)
```



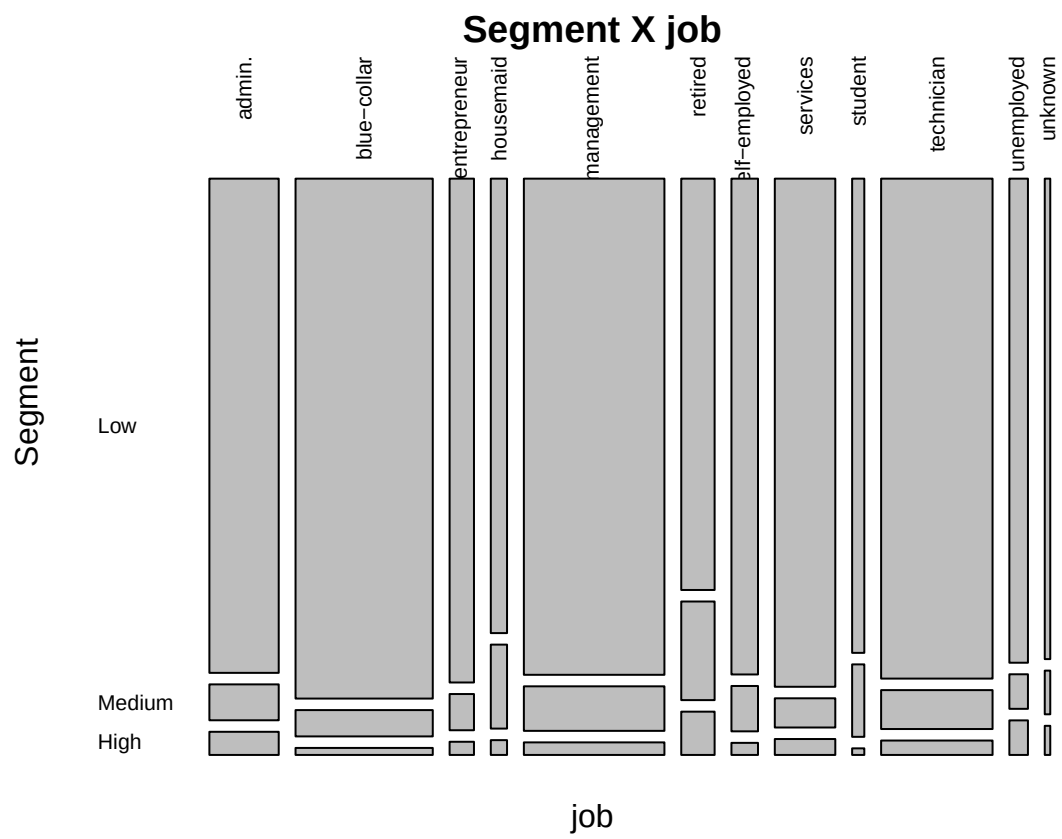
The population is primarily in *Low* segment, while *Medium* and *High* segments have a lot less samples, but more than a half of them are labelled positively.

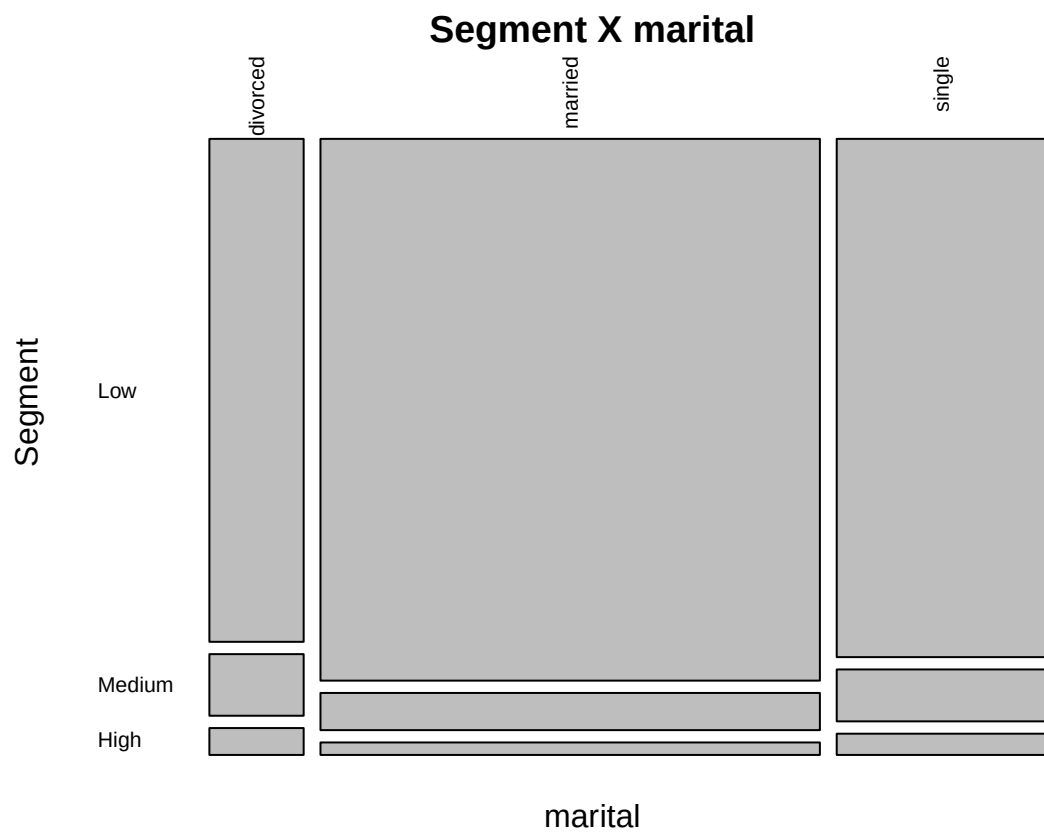
1.2 Data analysis

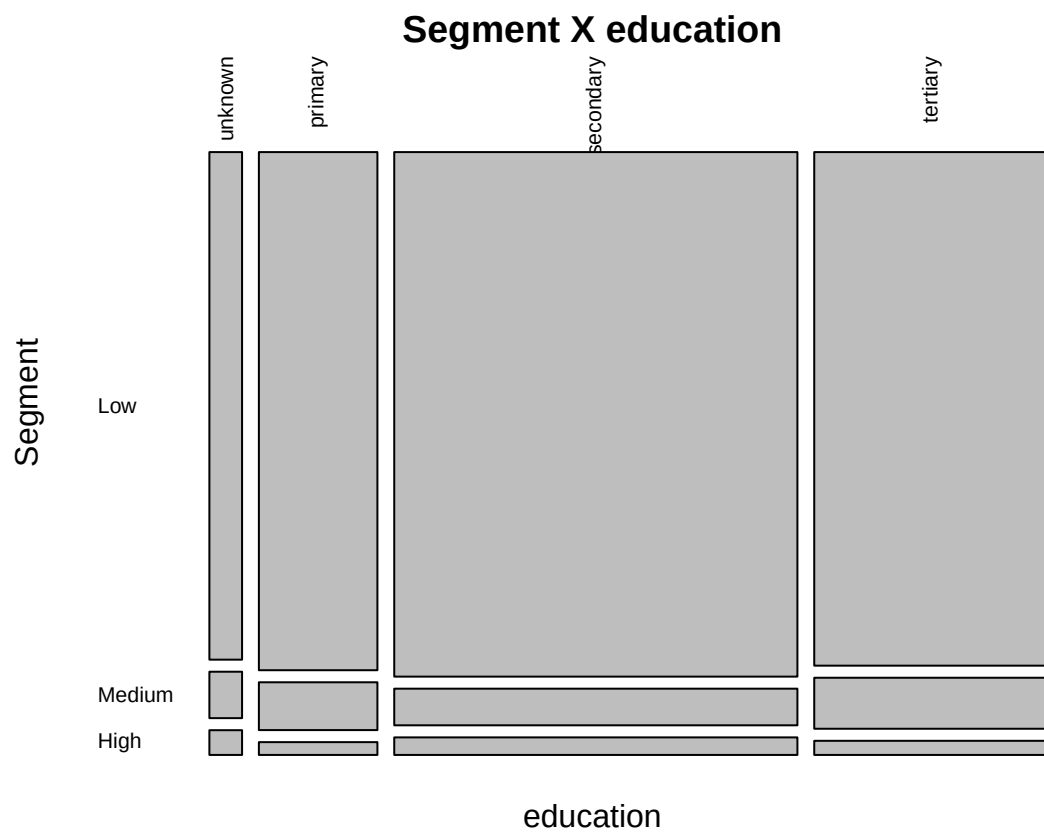
In this section we will conduct an examination of interactions between features and **segment**. It must generally follow structure of target variables interactions, though we expect to understand structure of data better.

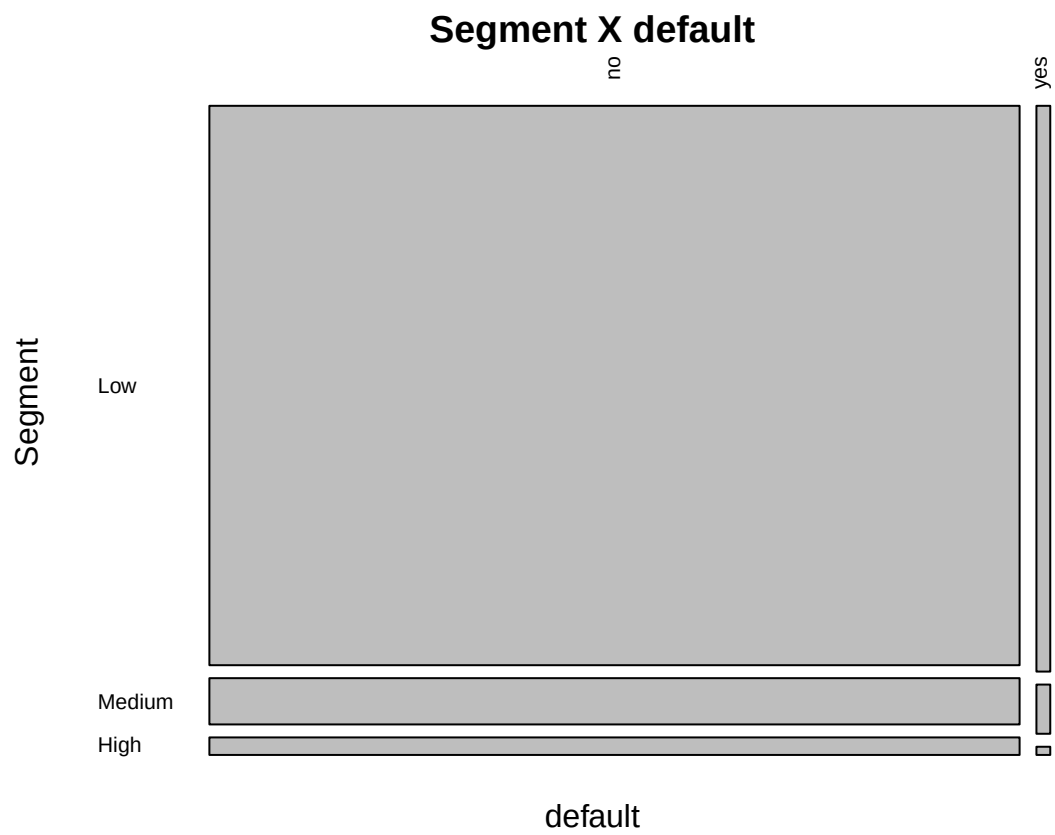
We will start by plotting mosaic plots of contingency tables segment versus categorical features.

```
for(col in colnames(data[sapply(data, is.factor)])) {
  if(col == "segment" || col == "y"){ next}
  ct = table(data[,col], data.joint_job$segment)
  par(mar=c(3,4,1,2))
  mosaicplot(
    ct, ylab="Segment", xlab=col,
    main=sprintf("Segment X %s", col),
    las = 2
  )
}
```



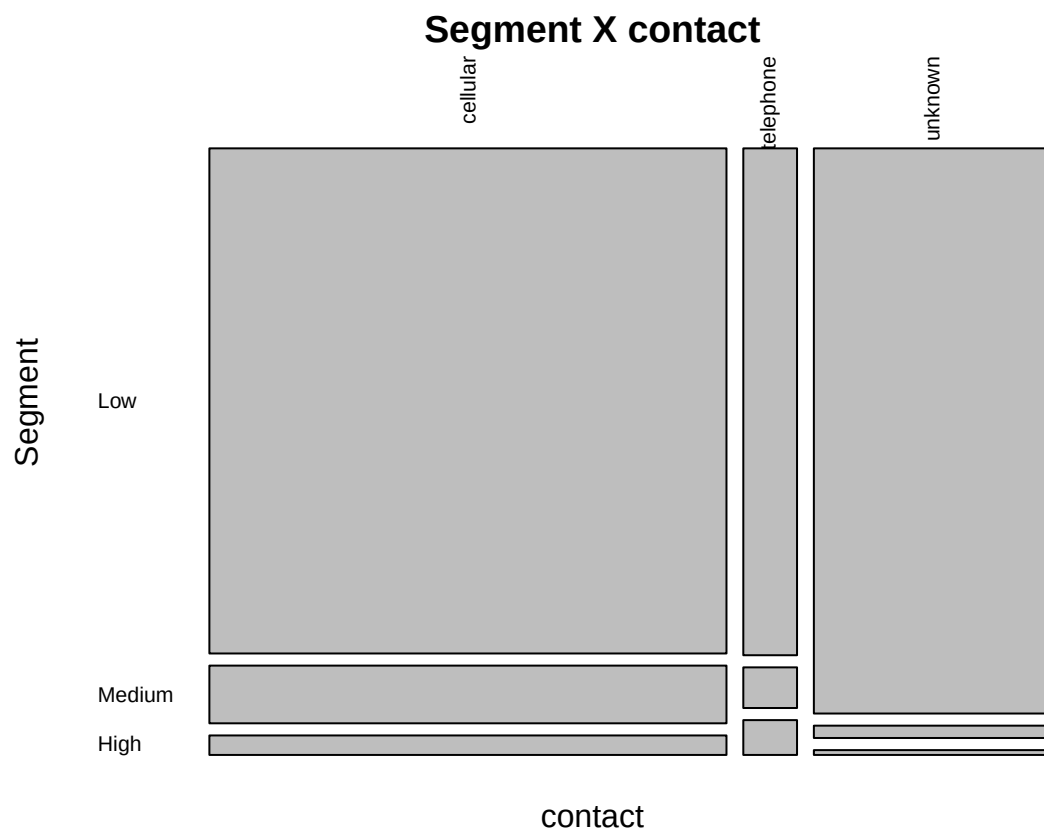


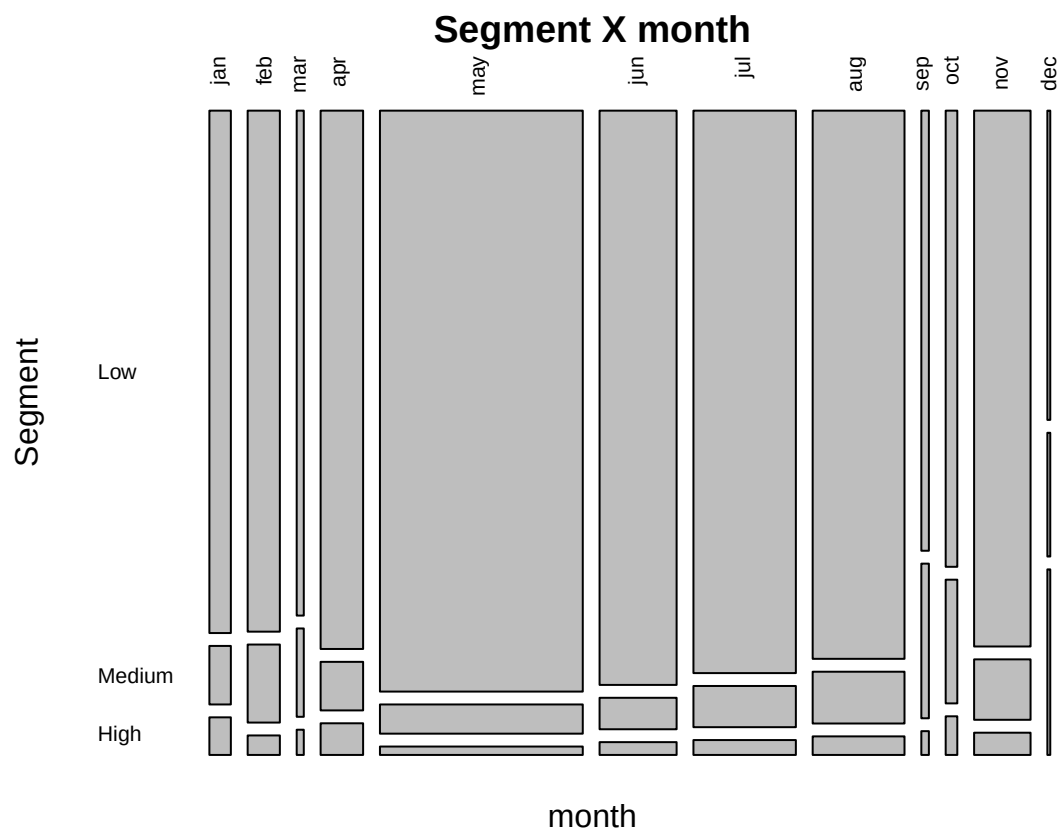


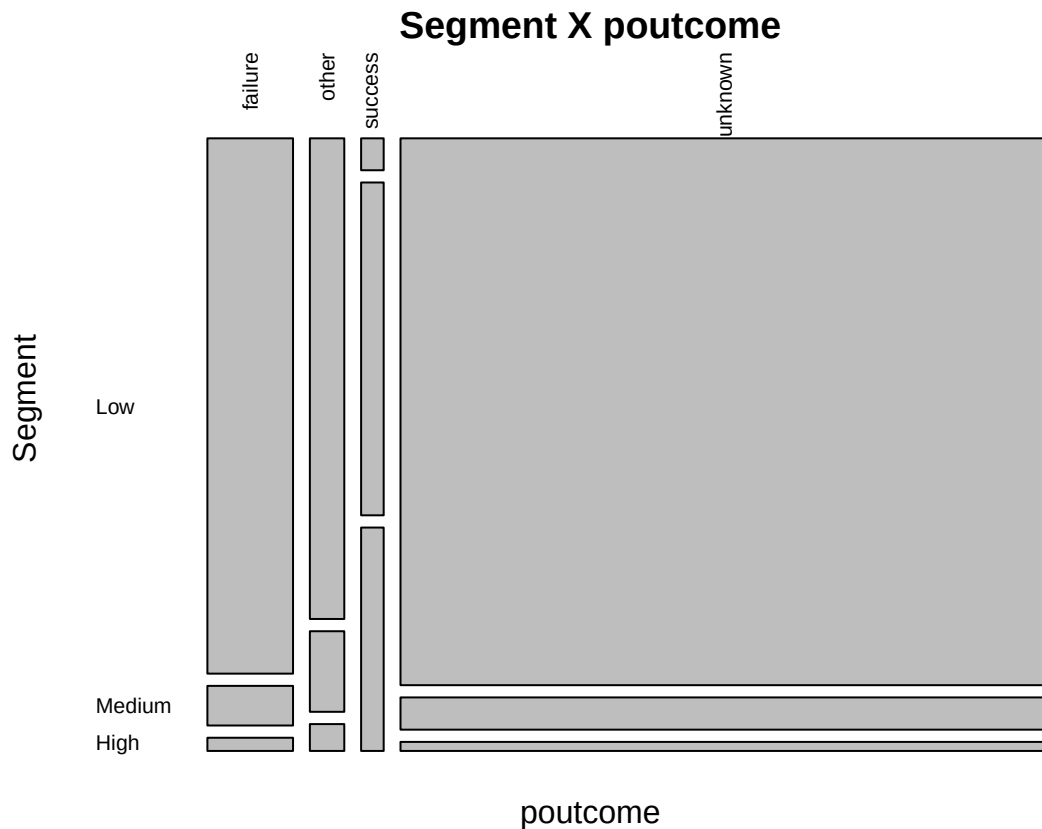








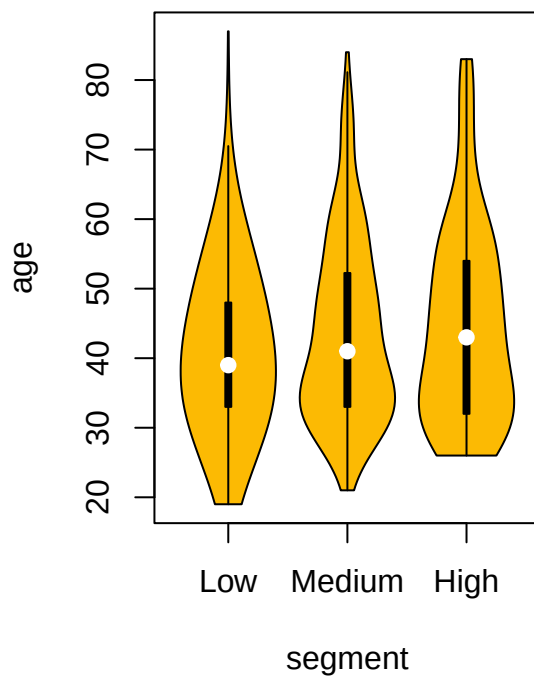
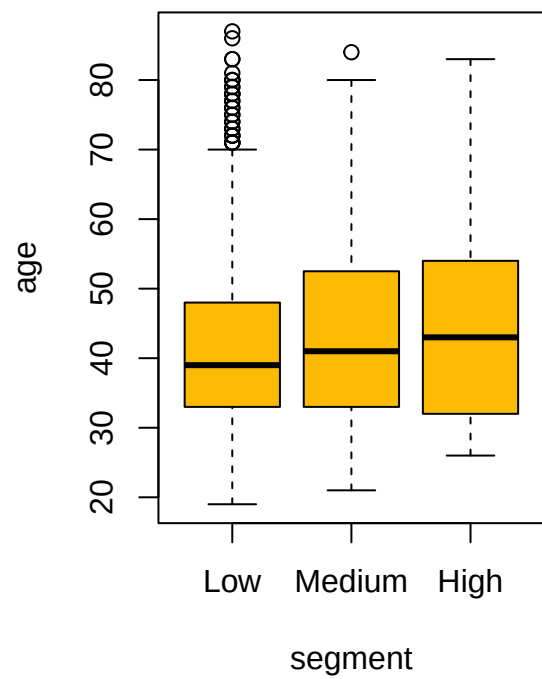


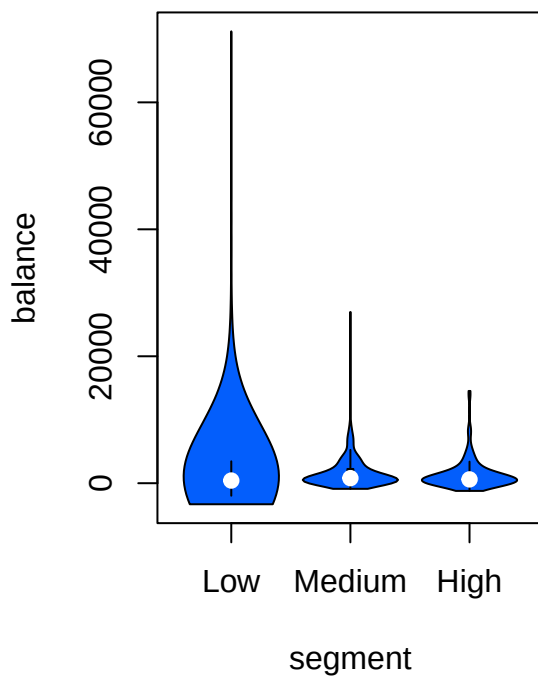
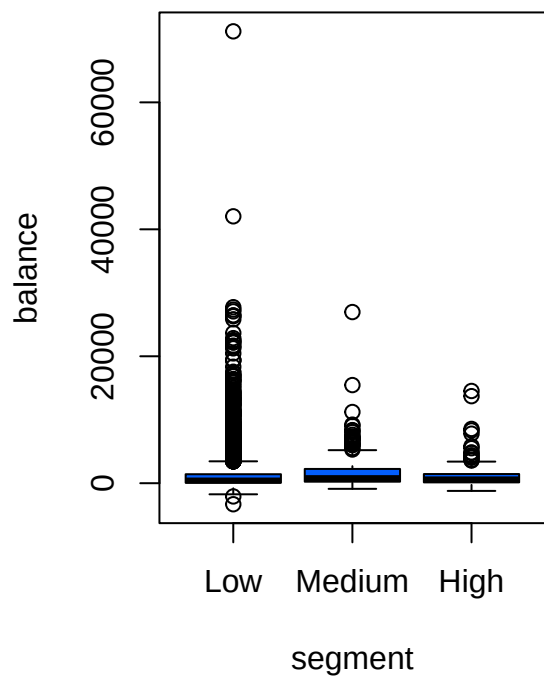


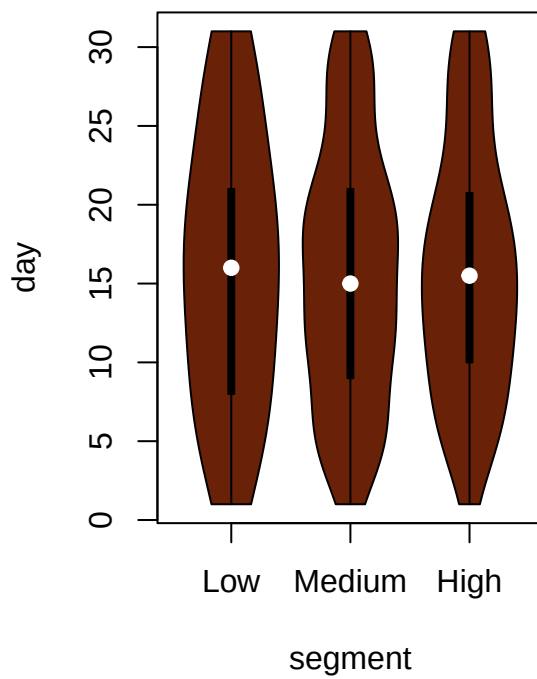
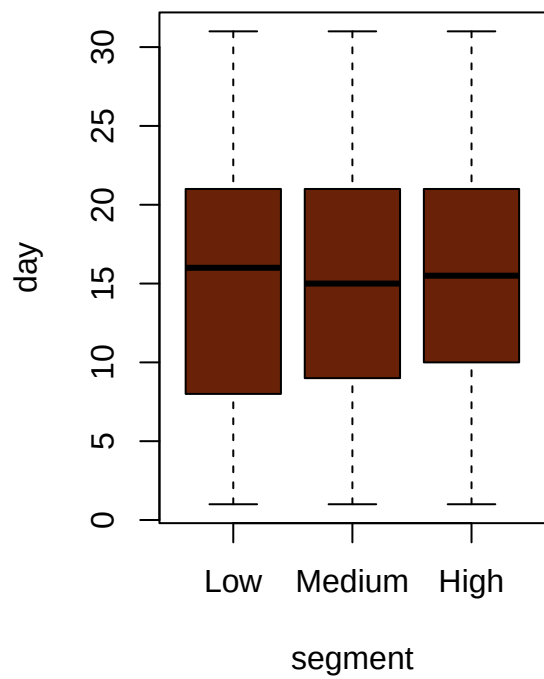
Most features do not differentiate between segments, the ones that do are **job** (specifically *retired*, *student* and *housemaid* are more likely to subscribe to a deposit), **housing** and **loan** (in both categories **no* is preferred by *Medium* segment) and **poutcome**** with *success* and *other* showing great proportion in *Medium* and *High*.

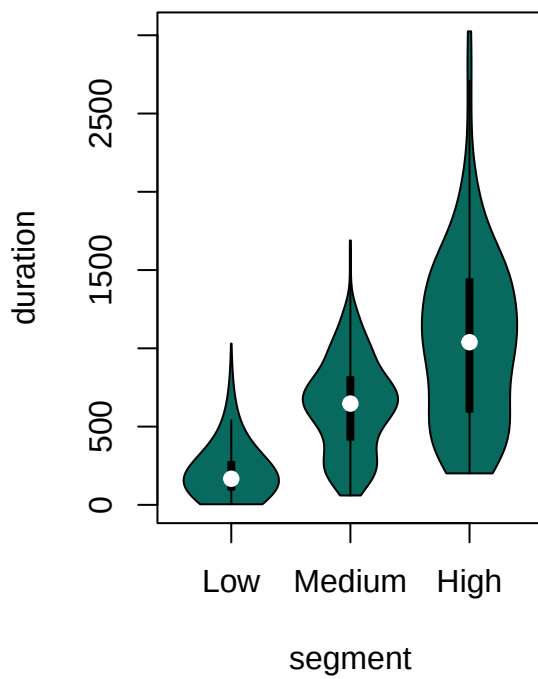
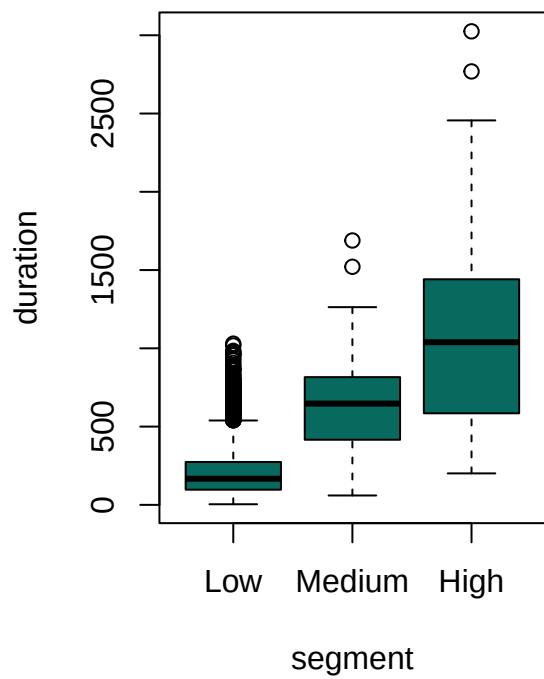
Now we will investigate relationship between **segment** and numerical features. We will use boxplot and violinplot.

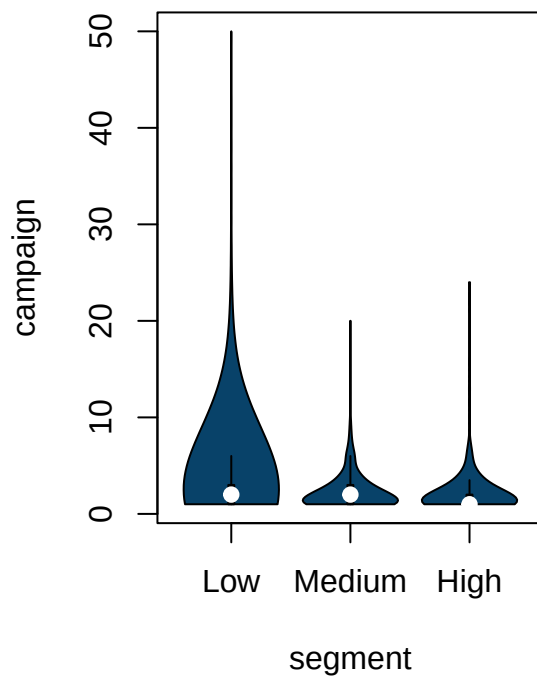
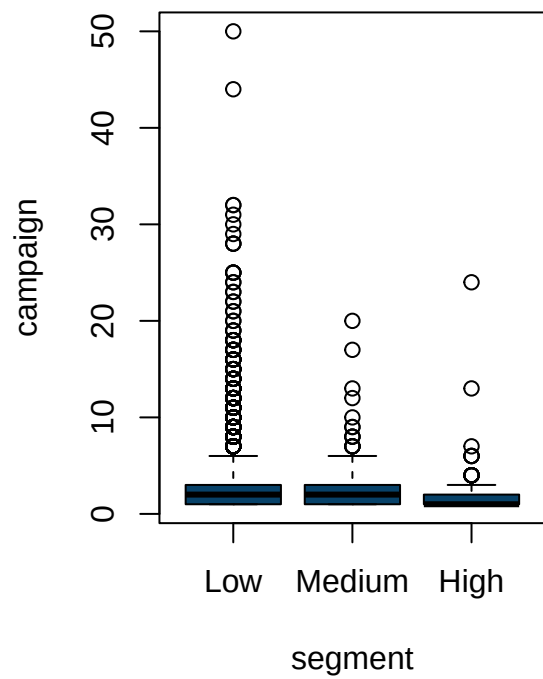
```
ylim=c(0, 1)
for(col in colnames(data.joint_job[apply(data.joint_job, is.numeric)])) {
  if(col == "score") {next}
  fml <- as.formula(paste(col, " ~ segment"))
  par(mfrow=c(1, 2))
  boxplot(fml, data.joint_job, col=COLORS[col])
  violplot(fml, data.joint_job, col=COLORS[col])
}
```

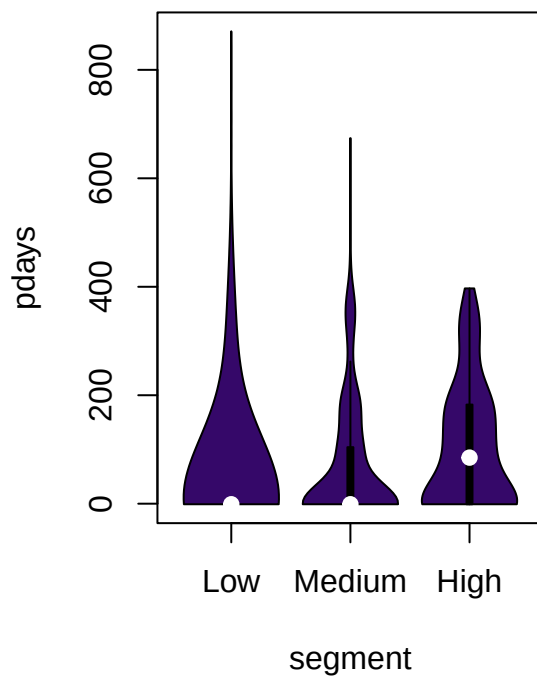
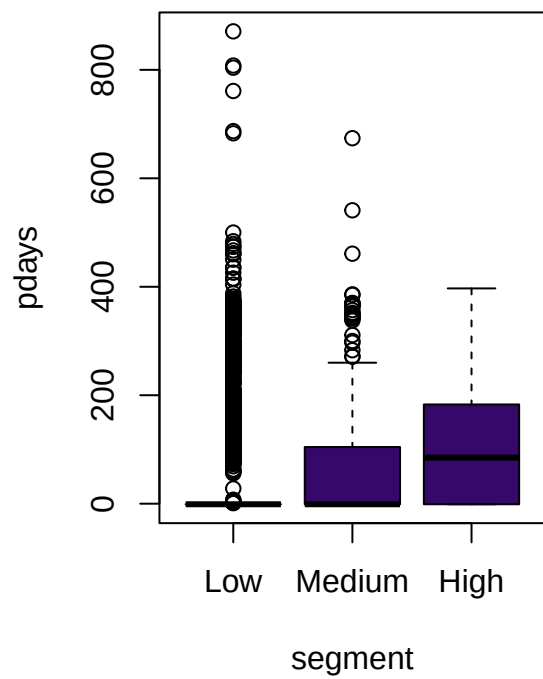


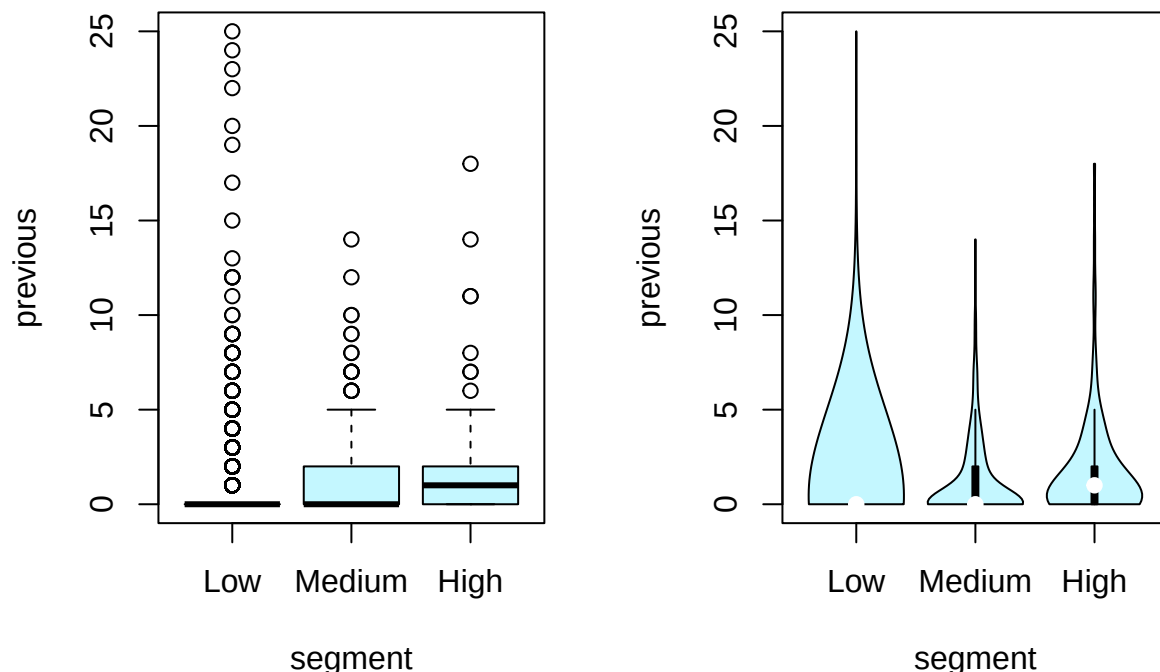












The greatest distinction is in **duration**, though as we have discussed it must be instrumental for subscribing to deposits. Minor interactions are present in **pdays**, **previous** and **balance** with *Low* values being concentrated in the neighborhood of zero. We will check those features for profiling.

1.3 Creating Profiles

We have gathered a packet of features and values to check: **housing** and **loan** with *no*; **job** with *retired*, *student* and *housemaid*; **pdays**, **previous** and **balance** with strictly positive values. We will create univariate and bivariate profiles, that balance between positive rate and coverage.

We will compute size of profiles, positive proportion. We will also conduct proportion test to check that deposit subscribers are more likely in the profile, than in its compliment, as well as a binomial test to check profile against the whole dataset. To measure how practically significant is profile we will compute Cohen's h, Risk Difference and Odds Ratio.

```
num_y <- ifelse(data$y == "yes", 1, 0)
cohen.h <- function(p1, p2) {
  2 * asin(sqrt(p1)) - 2 * asin(sqrt(p2))
}
analyse_profile <- function(profile_bool, baseline.data = num_y) {
  num_profile <- ifelse(baseline.data[profile_bool], 1, 0)
  num_comp_profile <- ifelse(baseline.data[!profile_bool], 1, 0)

  print("-----", quote=F)
  print(sprintf("Coverage:          %d samples = %.1f%% of dataset", length(num_profile), 100 * length(
  print(
    sprintf(
      "Positives:          %d samples = %.1f%% of profile = %.1f%% of all positives",
```

```

        sum(num_profile), 100 * mean(num_profile), 100 * sum(num_profile) / sum(baseline.data)
    ),
    quote=F
)
print("-----", quote=F)
if (length(num_profile) < 1 || sum(num_profile) < 1) return()

p_profile <- mean(num_profile)
p_baseline <- mean(baseline.data)

prop_pvalue <- prop.test(x = c(sum(num_profile), sum(num_comp_profile)), n = c(length(num_profile),
binom_pvalue <- binom.test(sum(num_profile), length(num_profile), p=p_baseline, alternative = "less
print(sprintf("Proportion test p-value against profile's compliment:      %.3f", prop_pvalue), quote=F)
print(sprintf("Exact Binomial test p-value:                               %.3f", binom_pvalue), quote=F)

print("-----", quote=F)
cohen_h <- cohen.h(p_profile, p_baseline)
risk_difference <- p_profile - p_baseline
odds_ratio <- p_profile * (1 - p_profile) / (p_baseline * (1 - p_baseline))

print(sprintf("Cohen's h value: %.3f", cohen_h), quote=F)
print(sprintf("Risk difference: %.3f", risk_difference), quote=F)
print(sprintf("Odds ratio:      %.3f", odds_ratio), quote=F)
print("-----", quote=F)
print("", quote=F)
}

```

1.3.1 Univariate

During exploration of single variable we would like to find profiles that have both wide coverage, have greater proportion of positive samples than dataset and have Cohen's h greater than 0.1.

```

print("Housing = no")

## [1] "Housing = no"
analyse_profile(data$housing == "no")

## [1] -----
## [1] Coverage:      1962 samples = 43.4% of dataset
## [1] Positives:     301 samples = 15.3% of profile = 57.8% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                               1.000
## [1] -----
## [1] Cohen's h value: 0.112
## [1] Risk difference: 0.038
## [1] Odds ratio:      1.274
## [1] -----
## [1]

print("Loan = no")

## [1] "Loan = no"

```

```
analyse_profile(data$loan == "no")
```

```
## [1] -----
## [1] Coverage:      3830 samples = 84.7% of dataset
## [1] Positives:      478 samples = 12.5% of profile = 91.7% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                0.969
## [1] -----
## [1] Cohen's h value: 0.029
## [1] Risk difference: 0.010
## [1] Odds ratio:      1.071
## [1] -----
## [1]
```

Both binary features show very wide coverage (housing - 43%, loan - 85%), though effect size metrics show low values. We will use these features in combinations with other features to try to improve positive rate.

```
print("Job = student")
```

```
## [1] "Job = student"
```

```
analyse_profile(data$job == "student")
```

```
## [1] -----
## [1] Coverage:      84 samples = 1.9% of dataset
## [1] Positives:      19 samples = 22.6% of profile = 3.6% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      0.999
## [1] Exact Binomial test p-value:                                0.999
## [1] -----
## [1] Cohen's h value: 0.299
## [1] Risk difference: 0.111
## [1] Odds ratio:      1.717
## [1] -----
## [1]
```

```
print("Job = housemaid")
```

```
## [1] "Job = housemaid"
```

```
analyse_profile(data$job == "housemaid")
```

```
## [1] -----
## [1] Coverage:      112 samples = 2.5% of dataset
## [1] Positives:      14 samples = 12.5% of profile = 2.7% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      0.571
## [1] Exact Binomial test p-value:                                0.692
## [1] -----
## [1] Cohen's h value: 0.030
## [1] Risk difference: 0.010
## [1] Odds ratio:      1.073
## [1] -----
## [1]
```

```
print("Job = retired")
```

```
## [1] "Job = retired"
```

```
analyse_profile(data$job == "retired")
```

```
## [1] -----
## [1] Coverage:      230 samples = 5.1% of dataset
## [1] Positives:      54 samples = 23.5% of profile = 10.4% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.319
## [1] Risk difference: 0.120
## [1] Odds ratio:      1.762
## [1] -----
## [1]
```

Both *student* and *housemaid* have awful coverage under 2.5%. *Housemaid* also has tiny improvement, so we discard it right away. *Retired* is a relatively large population with good positive rate, so we expect it to behave great in interaction profiles.

```
print("Previous outcome = success")
```

```
## [1] "Previous outcome = success"
```

```
analyse_profile(data$poutcome == "success")
```

```
## [1] -----
## [1] Coverage:      129 samples = 2.9% of dataset
## [1] Positives:      83 samples = 64.3% of profile = 15.9% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 1.169
## [1] Risk difference: 0.528
## [1] Odds ratio:      2.250
## [1] -----
## [1]
```

```
print("Previous outcome = other")
```

```
## [1] "Previous outcome = other"
```

```
analyse_profile(data$poutcome == "other")
```

```
## [1] -----
## [1] Coverage:      197 samples = 4.4% of dataset
## [1] Positives:      38 samples = 19.3% of profile = 7.3% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                0.999
## [1] -----
## [1] Cohen's h value: 0.217
## [1] Risk difference: 0.078
## [1] Odds ratio:      1.527
```

```
## [1] -----
## [1]
print("Previous outcome = other or success")

## [1] "Previous outcome = other or success"
analyse_profile((data$poutcome == "success") | (data$poutcome == "other"))

## [1] -----
## [1] Coverage:          326 samples = 7.2% of dataset
## [1] Positives:         121 samples = 37.1% of profile = 23.2% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                               1.000
## [1] -----
## [1] Cohen's h value: 0.617
## [1] Risk difference: 0.256
## [1] Odds ratio:       2.289
## [1] -----
## [1]
```

Previous outcome indeed shows great results with unified group covering 7.2% dataset and more than doubling the odds. The question is whether it is just a connection with a bank, or the result of previous campaign that let people to subscribe to a deposit.

```
print("Contacts during previous campaign > 0")

## [1] "Contacts during previous campaign > 0"
analyse_profile(data$previous > 0)

## [1] -----
## [1] Coverage:          816 samples = 18.0% of dataset
## [1] Positives:         184 samples = 22.5% of profile = 35.3% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                               1.000
## [1] -----
## [1] Cohen's h value: 0.297
## [1] Risk difference: 0.110
## [1] Odds ratio:       1.713
## [1] -----
## [1]
```

```
print("Contacts during previous campaign > 2")

## [1] "Contacts during previous campaign > 2"
analyse_profile(data$previous > 1)

## [1] -----
## [1] Coverage:          530 samples = 11.7% of dataset
## [1] Positives:         133 samples = 25.1% of profile = 25.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                               1.000
## [1] -----
## [1] Cohen's h value: 0.357
```

```
## [1] Risk difference: 0.136
## [1] Odds ratio:      1.844
## [1] -----
## [1]
```

```
print("Contacts during previous campaign > 5")
```

```
## [1] "Contacts during previous campaign > 5"
```

```
analyse_profile(data$pdays > 5)
```

```
## [1] -----
## [1] Coverage:      805 samples = 17.8% of dataset
## [1] Positives:     182 samples = 22.6% of profile = 34.9% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.298
## [1] Risk difference: 0.111
## [1] Odds ratio:      1.716
## [1] -----
## [1]
```

Shrinking allowed window only made the profile narrow, without improving positive rate, so we will stick with any strictly positive number of contacts during previous campaign.

```
print("Days since last conctact during previous campaign > 0")
```

```
## [1] "Days since last conctact during previous campaign > 0"
```

```
analyse_profile(data$pdays > 0)
```

```
## [1] -----
## [1] Coverage:      816 samples = 18.0% of dataset
## [1] Positives:     184 samples = 22.5% of profile = 35.3% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.297
## [1] Risk difference: 0.110
## [1] Odds ratio:      1.713
## [1] -----
## [1]
```

```
print("Days since last conctact during previous campaign > 100")
```

```
## [1] "Days since last conctact during previous campaign > 100"
```

```
analyse_profile(data$pdays > 100)
```

```
## [1] -----
## [1] Coverage:      680 samples = 15.0% of dataset
## [1] Positives:     122 samples = 17.9% of profile = 23.4% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
```

```
## [1] Cohen's h value: 0.182
## [1] Risk difference: 0.064
## [1] Odds ratio:      1.444
## [1] -----
## [1]
```

```
print("Days since last conctact during previous campaign > 200")
```

```
## [1] "Days since last conctact during previous campaign > 200"
```

```
analyse_profile(data$pdays > 200)
```

```
## [1] -----
## [1] Coverage:      382 samples = 8.4% of dataset
## [1] Positives:     53 samples = 13.9% of profile = 10.2% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:    0.922
## [1] Exact Binomial test p-value:                            0.933
## [1] -----
## [1] Cohen's h value: 0.071
## [1] Risk difference: 0.024
## [1] Odds ratio:     1.172
## [1] -----
## [1]
```

Pushing the boundary up for the number of days since last contact during previous campaign has made the profile only worse, so we will keep the widest that is *pdays* > 0.

```
print("Balance greater than 0")
```

```
## [1] "Balance greater than 0"
```

```
analyse_profile(data$balance > 0)
```

```
## [1] -----
## [1] Coverage:      3798 samples = 84.0% of dataset
## [1] Positives:     461 samples = 12.1% of profile = 88.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:    0.998
## [1] Exact Binomial test p-value:                            0.886
## [1] -----
## [1] Cohen's h value: 0.019
## [1] Risk difference: 0.006
## [1] Odds ratio:     1.046
## [1] -----
## [1]
```

```
print("Balance greater than 100")
```

```
## [1] "Balance greater than 100"
```

```
analyse_profile(data$balance > 100)
```

```
## [1] -----
## [1] Coverage:      3265 samples = 72.2% of dataset
## [1] Positives:     415 samples = 12.7% of profile = 79.7% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:    1.000
## [1] Exact Binomial test p-value:                            0.983
```

```

## [1] -----
## [1] Cohen's h value: 0.036
## [1] Risk difference: 0.012
## [1] Odds ratio:      1.088
## [1] -----
## [1]

print("Balance greater than 500")

## [1] "Balance greater than 500"

analyse_profile(data$balance > 500)

## [1] -----
## [1] Coverage:      2134 samples = 47.2% of dataset
## [1] Positives:      302 samples = 14.2% of profile = 58.0% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.079
## [1] Risk difference: 0.026
## [1] Odds ratio:      1.192
## [1] -----
## [1]

print("Balance greater than 1000")

## [1] "Balance greater than 1000"

analyse_profile(data$balance > 1000)

## [1] -----
## [1] Coverage:      1481 samples = 32.8% of dataset
## [1] Positives:      221 samples = 14.9% of profile = 42.4% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.101
## [1] Risk difference: 0.034
## [1] Odds ratio:      1.245
## [1] -----
## [1]

print("Balance greater than 5000")

## [1] "Balance greater than 5000"

analyse_profile(data$balance > 5000)

## [1] -----
## [1] Coverage:      309 samples = 6.8% of dataset
## [1] Positives:      34 samples = 11.0% of profile = 6.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      0.419
## [1] Exact Binomial test p-value:                                0.430
## [1] -----

```

```
## [1] Cohen's h value: -0.016
## [1] Risk difference: -0.005
## [1] Odds ratio:      0.960
## [1] -----
## [1]
```

Balance over one thousand has great coverage of 33% and a promising positive rate of 15%. We hope to improve with other features.

1.3.2 Bivariate

Here we will try to refine our profiles.

```
print("Previous outcome = success & housing = no")
```

```
## [1] "Previous outcome = success & housing = no"
```

```
analyse_profile((data$poutcome == "success") & (data$housing == "no"))
```

```
## [1] -----
## [1] Coverage:      89 samples = 2.0% of dataset
## [1] Positives:     63 samples = 70.8% of profile = 12.1% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 1.307
## [1] Risk difference: 0.593
## [1] Odds ratio:      2.028
## [1] -----
## [1]
```

```
print("Previous outcome = success & loan = no")
```

```
## [1] "Previous outcome = success & loan = no"
```

```
analyse_profile((data$poutcome == "success") & (data$loan == "no"))
```

```
## [1] -----
## [1] Coverage:      123 samples = 2.7% of dataset
## [1] Positives:     78 samples = 63.4% of profile = 15.0% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 1.150
## [1] Risk difference: 0.519
## [1] Odds ratio:      2.275
## [1] -----
## [1]
```

Intersection with **poutcome** covers a small patch of dataset, so we will not proceed with this.

```
print("Previous outcome = other & housing = no")
```

```
## [1] "Previous outcome = other & housing = no"
```

```
analyse_profile((data$poutcome == "other") & (data$housing == "no"))
```

```
## [1] -----
```

```
## [1] Coverage:          64 samples = 1.4% of dataset
## [1] Positives:          17 samples = 26.6% of profile = 3.3% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.390
## [1] Risk difference: 0.150
## [1] Odds ratio:       1.913
## [1] -----
## [1]
```

```
print("Previous outcome = other & loan = no")
```

```
## [1] "Previous outcome = other & loan = no"
```

```
analyse_profile((data$poutcome == "other") & (data$loan == "no"))
```

```
## [1] -----
## [1] Coverage:          173 samples = 3.8% of dataset
## [1] Positives:          34 samples = 19.7% of profile = 6.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                0.999
## [1] -----
## [1] Cohen's h value: 0.226
## [1] Risk difference: 0.081
## [1] Odds ratio:       1.549
## [1] -----
## [1]
```

poutcome *other* profiles are also very small and additionally not as good.

```
print("Previous outcome = (other or success) & housing = no")
```

```
## [1] "Previous outcome = (other or success) & housing = no"
```

```
analyse_profile((data$poutcome %in% c("other", "success")) & (data$housing == "no"))
```

```
## [1] -----
## [1] Coverage:          153 samples = 3.4% of dataset
## [1] Positives:          80 samples = 52.3% of profile = 15.4% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.924
## [1] Risk difference: 0.408
## [1] Odds ratio:       2.447
## [1] -----
## [1]
```

```
print("Previous outcome = (other or success) & loan = no")
```

```
## [1] "Previous outcome = (other or success) & loan = no"
```

```
analyse_profile((data$poutcome %in% c("other", "success")) & (data$loan == "no"))
```

```
## [1] -----
```

```
## [1] Coverage:          296 samples = 6.5% of dataset
## [1] Positives:          112 samples = 37.8% of profile = 21.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.632
## [1] Risk difference: 0.263
## [1] Odds ratio:       2.307
## [1] -----
## [1]
```

Union of previous profiles gives better results, but still covers little patch.

```
print("Job = retired & housing = no")
```

```
## [1] "Job = retired & housing = no"
```

```
analyse_profile((data$job == "retired") & (data$housing == "no"))
```

```
## [1] -----
## [1] Coverage:          180 samples = 4.0% of dataset
## [1] Positives:          47 samples = 26.1% of profile = 9.0% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.380
## [1] Risk difference: 0.146
## [1] Odds ratio:       1.892
## [1] -----
## [1]
```

```
print("Job = retired & loan = no")
```

```
## [1] "Job = retired & loan = no"
```

```
analyse_profile((data$job == "retired") & (data$loan == "no"))
```

```
## [1] -----
## [1] Coverage:          198 samples = 4.4% of dataset
## [1] Positives:          52 samples = 26.3% of profile = 10.0% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.383
## [1] Risk difference: 0.147
## [1] Odds ratio:       1.899
## [1] -----
## [1]
```

Those profiles are good, though coverage of 4% is not ideal, let us seek further.

```
print("Job = student & housing = no")
```

```
## [1] "Job = student & housing = no"
```

```
analyse_profile((data$job == "student") & (data$housing == "no"))
```

```
## [1] -----
## [1] Coverage:          64 samples = 1.4% of dataset
## [1] Positives:          17 samples = 26.6% of profile = 3.3% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.390
## [1] Risk difference: 0.150
## [1] Odds ratio:       1.913
## [1] -----
## [1]
```

```
print("Job = student & loan = no")
```

```
## [1] "Job = student & loan = no"
```

```
analyse_profile((data$job == "student") & (data$loan == "no"))
```

```
## [1] -----
## [1] Coverage:          83 samples = 1.8% of dataset
## [1] Positives:          19 samples = 22.9% of profile = 3.6% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      0.999
## [1] Exact Binomial test p-value:                                0.999
## [1] -----
## [1] Cohen's h value: 0.305
## [1] Risk difference: 0.114
## [1] Odds ratio:       1.731
## [1] -----
## [1]
```

Student profiles was refined only into even smaller profile, so we will reject this profile.

```
analyse_profile((data$pdays > 0) & ((data$poutcome == "other") | (data$poutcome == "success")))
```

```
## [1] -----
## [1] Coverage:          326 samples = 7.2% of dataset
## [1] Positives:          121 samples = 37.1% of profile = 23.2% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.617
## [1] Risk difference: 0.256
## [1] Odds ratio:       2.289
## [1] -----
## [1]
```

Unsurprisingly this profile matches **poutcome** *other* or *success* completely, this was done more as an integrity check.

```
analyse_profile((data$loan == "no") | (data$housing == "no"))
```

```
## [1] -----
## [1] Coverage:          4115 samples = 91.0% of dataset
```

```
## [1] Positives:          496 samples = 12.1% of profile = 95.2% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                0.861
## [1] -----
## [1] Cohen's h value: 0.016
## [1] Risk difference: 0.005
## [1] Odds ratio:       1.040
## [1] -----
## [1]
```

Union of two of the widest profiles has resulted in the profile that spans more than 90% of data, but positive rate is close to the baseline, so we will not use this profile.

```
analyse_profile((data$job == "retired") & (data$balance > 0))
```

```
## [1] -----
## [1] Coverage:          200 samples = 4.4% of dataset
## [1] Positives:          47 samples = 23.5% of profile = 9.0% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.319
## [1] Risk difference: 0.120
## [1] Odds ratio:       1.763
## [1] -----
## [1]
```

This is not the best profile, but it will be easy to search for such people so, we are going to use it later.

```
analyse_profile((data$housing == "no") & (data$pdays > 0))
```

```
## [1] -----
## [1] Coverage:          300 samples = 6.6% of dataset
## [1] Positives:          107 samples = 35.7% of profile = 20.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.587
## [1] Risk difference: 0.241
## [1] Odds ratio:       2.250
## [1] -----
## [1]
```

This profile is extremely good as it shows coverage of 6.6% while more than its third had subscribed to the deposit.

1.4 Profiles

We have chosen three profiles:

- Previous Campaign Outcome = other or success (have good history with the bank)
- Housing = no and Previous days > 0 (have being previously contacted and have capacity to subscribe to a deposit)
- Job = retired and balance > 0 (retired people who have capacity and motivation to grow their fortune)

Let us take a look at individual profiles once more and then investigate how different they really are.

```
print("Previous Campaign Outcome = other or success")
```

```
## [1] "Previous Campaign Outcome = other or success"
```

```
analyse_profile(data$poutcome %in% c("other", "success"))
```

```
## [1] -----
## [1] Coverage:      326 samples = 7.2% of dataset
## [1] Positives:     121 samples = 37.1% of profile = 23.2% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:    1.000
## [1] Exact Binomial test p-value:                            1.000
## [1] -----
## [1] Cohen's h value: 0.617
## [1] Risk difference: 0.256
## [1] Odds ratio:      2.289
## [1] -----
## [1]
```

```
print("Housing = no and Previous days > 0")
```

```
## [1] "Housing = no and Previous days > 0"
```

```
analyse_profile((data$housing == "no") & (data$pdays > 0))
```

```
## [1] -----
## [1] Coverage:      300 samples = 6.6% of dataset
## [1] Positives:     107 samples = 35.7% of profile = 20.5% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:    1.000
## [1] Exact Binomial test p-value:                            1.000
## [1] -----
## [1] Cohen's h value: 0.587
## [1] Risk difference: 0.241
## [1] Odds ratio:      2.250
## [1] -----
## [1]
```

```
print("Job = retired and balance > 0")
```

```
## [1] "Job = retired and balance > 0"
```

```
analyse_profile((data$job == "retired") & (data$balance > 0))
```

```
## [1] -----
## [1] Coverage:      200 samples = 4.4% of dataset
## [1] Positives:      47 samples = 23.5% of profile = 9.0% of all positives
## [1] -----
## [1] Proportion test p-value against profile's compliment:    1.000
## [1] Exact Binomial test p-value:                            1.000
## [1] -----
## [1] Cohen's h value: 0.319
## [1] Risk difference: 0.120
## [1] Odds ratio:      1.763
## [1] -----
## [1]
```

Now we will take a look at pairwise intersections of profiles

```
print("Previous Campaign Outcome = other or success and Housing = no and Previous days > 0")
```

```
## [1] "Previous Campaign Outcome = other or success and Housing = no and Previous days > 0"
```

```
prof <- data$poutcome %in% c("other", "success") & (data$housing == "no") & (data$pdays > 0)
print(sprintf("Number of samples: %d = %.1f%% of dataset", sum(prof), 100 * mean(prof)))
```

```
## [1] "Number of samples: 153 = 3.4% of dataset"
```

```
print("", quote=F)
```

```
## [1]
```

```
print("Housing = no and Previous days > 0 and Job = retired and balance > 0")
```

```
## [1] "Housing = no and Previous days > 0 and Job = retired and balance > 0"
```

```
prof <- (data$housing == "no") & (data$pdays > 0) & (data$job == "retired") & (data$balance > 0)
print(sprintf("Number of samples: %d = %.1f%% of dataset", sum(prof), 100 * mean(prof)))
```

```
## [1] "Number of samples: 40 = 0.9% of dataset"
```

```
print("", quote=F)
```

```
## [1]
```

```
print("Job = retired and balance > 0 and Previous Campaign Outcome = other or success")
```

```
## [1] "Job = retired and balance > 0 and Previous Campaign Outcome = other or success"
```

```
prof <- (data$job == "retired") & (data$balance > 0) & data$poutcome %in% c("other", "success")
print(sprintf("Number of samples: %d = %.1f%% of dataset", sum(prof), 100 * mean(prof)))
```

```
## [1] "Number of samples: 16 = 0.4% of dataset"
```

```
print("", quote=F)
```

```
## [1]
```

```
print("Intersection of all profiles")
```

```
## [1] "Intersection of all profiles"
```

```
prof <- (data$job == "retired") & (data$balance > 0) & data$poutcome %in% c("other", "success") & (data$housing == "no") & (data$pdays > 0)
print(sprintf("Number of samples: %d = %.1f%% of dataset", sum(prof), 100 * mean(prof)))
```

```
## [1] "Number of samples: 16 = 0.4% of dataset"
```

Most of intersections have neglectable intersection, but **poutcome** = *other* or *success* with **housing** = *no* and **pdays** > 0 intersect roughly as a half of their profiles. They still represent different populations. Let us see union of all profiles.

```
analyse_profile(
  (data$poutcome %in% c("other", "success")) |
  (data$housing == "no") & (data$pdays > 0) |
  (data$job == "retired") & (data$balance > 0)
)
```

```
## [1] -----
```

```
## [1] Coverage:      633 samples = 14.0% of dataset
```

```
## [1] Positives:    181 samples = 28.6% of profile = 34.7% of all positives
```

```
## [1] -----
## [1] Proportion test p-value against profile's compliment:      1.000
## [1] Exact Binomial test p-value:                                1.000
## [1] -----
## [1] Cohen's h value: 0.436
## [1] Risk difference: 0.171
## [1] Odds ratio:      2.003
## [1] -----
## [1]
```

Together profiles cover more than a third of all who had subscribed to the deposit, while covering 14% of population. It also doubles the odds of subscription compared to raw dataset. We call it a success.

1.5 Conclusion

We have done thorough examination of profiles and arrived at three profiles, that cover total 35% of people who subscribe. The room for improvement is to check other multivariate profiles, those could be guided by a different model, perhaps more flexible like random forest, polynomial logistic regression or support vector machine.